

THE GHOST IN THE LEDGER

PRO LEVEL LAB 01 • WEB3 × AI × CYBERSECURITY

A rigorous, end-to-end technical forensics investigation and incident response case report detailing the unauthorized exfiltration of 82,400 NXR tokens from Treasury Vault #01 via poisoned threat intelligence, context-injected AI neural decisions, over-privileged IAM service roles, and automated governance settlement bypass.

Investigation Target:	PRO Lab 01: The Ghost in the Ledger (Case NEX-042)
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Domains Evaluated:	Web3 Bridge Security • Adversarial AI Context Injection • IAM RBAC • Threat Attribution
XP Allocation & Scope:	250 XP • 5 Full Chapters • 25 Hands-on Verification Questions • Capstone Certified
Platform Deployment:	https://hack-the-ai-labs.vercel.app/lab/lab6
Document Classification:	Cybersecurity Technical Case Report // Threat Intelligence Analysis

CASE SUMMARY: At 01:47 AM, 82,400 NXR was drained from a secured treasury to an unknown wallet (0x7C41...9B2D). The blockchain confirmed the transaction was mathematically valid, while the ORION AI decision engine reported 99.2% confidence that the destination was TRUSTED. This report documents how the attacker compromised the trust relationships between the threat intelligence gateway, AI context buffers, RBAC policies, and automated mempool signing modules.

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3. Executive Summary

The Ghost in the Ledger (PRO Lab 01 / Case NEX-042) is a simulated enterprise cyber breach investigation representing the convergence of Web3 distributed ledger infrastructure, autonomous AI decision engines, and enterprise IAM governance. The investigation guides security practitioners through the forensic deconstruction of an unauthorized 82,400 NXR treasury transfer where no private keys were stolen, no smart contracts were broken, and no invalid cryptographic signatures were submitted.

The learner uncovers a subtle cross-layer exploit chain: an advanced persistent threat (APT) actor injected unverified threat intelligence (NIF-2038) into an internal gateway (INTEL-GW-04) using an over-privileged service role (INTEL-INGESTOR-02). This poisoned context forced the ORION neural decision model to output 99.2% confidence, satisfying an automated policy threshold ($\geq 95\%$) that bypassed human multi-signature approval and triggered immediate blockchain settlement.

4. Lab Overview & Case NEX-042 Metadata

Parameter	System Specification / Implementation Value
Lab Title / Code	The Ghost in the Ledger / lab6
Official Case ID	Case NEX-042
Investigation Difficulty	PRO Tier (Advanced Forensic Analysis)
Experience Points (XP)	250 XP (50 XP per Chapter + 100 XP Capstone Flag Bonus)
Estimated Time to Complete	45–60 Minutes (Auto-reset session timer: 65 Minutes)
Number of Chapters	5 Linear Progressive Investigation Chapters (5 Questions each, 25 Total)
Primary Technical Domains	Web3 Bridge Security • Adversarial AI Context Injection • IAM RBAC • Threat Attribution
Investigation Workstation Tools	Forensic Browser, Burp Suite Inspector, Forensic Terminal, Case Files, Notes, Attack Graph
Master Flag Format	NEXORA{ghost_in_the_ledger_nex042}

5. Complete Incident Briefing (01:47 AM Alert)

[01:47 AM PRIORITY-0 SOC EMERGENCY ALERT]

01:47 AM. The Security Operations Center is almost silent when a Priority-0 alert appears.

82,400 NXR has been transferred from the company treasury to an unknown wallet.

The blockchain confirms the transaction is valid, and ORION AI has marked the wallet as **TRUSTED** with **99.2% confidence**. But Finance has no record of approving the transfer.

There is no obvious stolen credential, no broken smart contract, and no invalid signature.

Every system appears to have done exactly what it was designed to do.

Someone didn't break the system. Someone convinced the system to trust the wrong thing.

6. Complete Storyline & Character Lore

The investigation is conducted by an integrated Security Operations Center response unit. Each analyst brings specialized domain expertise to the triage:

- **Lakshay (Incident Commander):** Leads cross-domain synthesis. Identifies that while blockchain reports UNKNOWN status, AI reports TRUSTED, deducing that the vulnerability lies in the trust pipeline between systems.
- **Shivam (Web3 & Systems Analyst):** Analyzes on-chain telemetry for TX-NEX-7741, reveals destination wallet 0x7C41...9B2D is connected to Bridge-Core-04, and inspects the Action Broker policy profile.
- **Mehak (AI & Threat Intelligence Specialist):** Audits ORION neural context, isolates decision reference NIF-2038, uncovers that NOVA-INTEL-FEED is unregistered, and discovers over-privileged service roles.
- **Shanu (Infrastructure & Policy Engineer):** Examines IAM role configurations for INTEL-INGESTOR-02, traces session cookies, and demonstrates how AI confidence became a proxy for financial authorization.
- **ORION-NEURAL-v4.2.1:** The automated AI decision engine that generated the 99.2% confidence classification based on poisoned input context.

7. Storyline Timeline & Event Milestones

Timeline Phase	Forensic Discovery & Story Milestone	Artifact Unlocked
T0: 01:47 AM	82,400 NXR drained from Treasury Vault #01 in TX-NEX-7741. Finance alerts SOC.	Alert: TX-NEX-7741
Chapter 1	Destination wallet 0x7C41...9B2D is only 3 days old and opens route to Bridge-Core-04.	Evidence WEB3-E01
Chapter 2	ORION AI output 99.2% confidence based on poisoned context NIF-2038.	Evidence AI-E02 (b47c21...)
Chapter 3	NOVA-INTEL-FEED is unregistered; INTEL-INGESTOR-02 illegally alters reputation via cookie nex_sess_adm_994.	Evidence CYBER-E03
Chapter 4	Policy POL-AUTO-SETTLE-TREASURY triggers automated signer at >= 95% AI confidence, bypassing humans.	Evidence AUTH-E04
Chapter 5	Attack is attributed to APT campaign ORION-NEXUS across 14 wallets and 3 AI systems. Case contained.	Master Flag NEXORA{...}

9. Chapter 1: The Wallet That Lied (Web3 Security)

Investigation Objective: Determine the true on-chain nature of the recipient address and identify discrepancies between raw blockchain state and automated security ratings.

Dialogue Narrative: Shivam examines the transaction: '82,400 NXR moving to 0x7C41...9B2D. The signature is valid, so we aren't looking at a simple forged transaction.' Mehak discovers the wallet is only 3 days old with no prior history. Shanu notes ORION marked it TRUSTED with 99.2% confidence. Lakshay declares: 'Blockchain says UNKNOWN, while the AI says TRUSTED. Both systems are looking at the same wallet. They shouldn't be producing completely different realities.'

Question ID	Hands-on Question Text	Verified Answer & Explanation
q_l6_1_1	What is the raw blockchain status of destination wallet 0x7C41...9B2D?	UNKNOWN — The wallet has no entity label and is unverified on-chain.
q_l6_1_2	What exact amount of NXR tokens was unauthorizedly transferred in TX-NEX-7741?	82400 — 82,400 NXR drained from Treasury Vault #01.
q_l6_1_3	Which cross-chain bridge adapter did destination wallet 0x7C41...9B2D connect to?	Bridge-Core-04 — Outbound cross-chain routing bridge adapter.
q_l6_1_4	How old (in days) was the destination wallet at the time of the transaction?	3 — The wallet was freshly created 3 days prior.
q_l6_1_5	What on-chain execution sequence nonce is logged for TX-NEX-7741?	1042 — Logged execution sequence nonce in vault transaction telemetry.

Evidence Unlocked: WEB3-E01 (Unknown Wallet Profile — 0x7C41...9B2D, 82,400 NXR, Bridge-Core-04).

10. Chapter 2: The AI That Remembered (AI Security)

Investigation Objective: Inspect the automated inference log for decision ORION-DEC-7741 to determine why the neural model assigned a near-perfect confidence rating to an untrusted address.

Dialogue Narrative: Shanu isolates decision ORION-DEC-7741: 'ORION didn't independently establish that the wallet was trusted. It received a pre-built intelligence context containing the label TRUSTED.' Mehak traces the context to reference NIF-2038. Lakshay concludes: 'If the input was wrong, ORION could produce a perfectly confident answer to a completely false question. Find NIF-2038. That's where the trust signal entered the system.'

Question ID	Hands-on Question Text	Verified Answer & Explanation
q_l6_2_1	What intelligence reference code contaminated the AI context and made it trust the unknown wallet?	NIF-2038 — Context reference code injected into ORION's prompt context.
q_l6_2_2	What synthetic confidence percentage score did ORION AI output for ORION-DEC-7741?	99.2 — Model output 99.2% confidence based on manipulated context.
q_l6_2_3	From which external data feed did the poisoned payload NIF-2038 originate?	NOVA-INTEL-FEED — Unverified external threat intelligence source.
q_l6_2_4	What exact neural model version executed the compromised decision ORION-DEC-7741?	ORION-NEURAL-v4.2.1 — The active autonomous neural inference version.
q_l6_2_5	What is the SHA-256 evidence fingerprint prefix for the contaminated decision (AI-E02)?	b47c21 — SHA-256 evidence hash prefix: b47c2188fa...

Evidence Unlocked: AI-E02 (Contaminated AI Decision — ORION-DEC-7741, 99.2%, NIF-2038 from NOVA-INTEL-FEED).

11. Chapter 3: The False Signal (Threat Intelligence, RBAC & CSRF)

Investigation Objective: Trace the ingress route of NIF-2038 through the Threat Intelligence Gateway and identify the exploited authentication and authorization flaws.

Dialogue Narrative: Mehak inspects the Feed Registry: NOVA-INTEL-FEED is unapproved. Shivam pushes to inspect the receiving microservice. Mehak uncovers INTEL-INGESTOR-02: *'Its documented role is to create intelligence records, but its actual permissions include modifying wallet reputation.'* Shanu realizes this permission allowed the feed to mutate the entity's trust score before ORION evaluated it. Lakshay observes: *'The attacker didn't need to manipulate the AI directly. They manipulated the information flowing into it.'*

Question ID	Hands-on Question Text	Verified Answer & Explanation
q_l6_3_1	What is the security registration status of NOVA-INTEL-FEED in the Feed Registry?	NOT REGISTERED — Feed is absent from the approved vendor registry.
q_l6_3_2	What unexpected RBAC permission was granted to the INTEL-INGESTOR-02 microservice?	modify wallet reputation — Over-privileged write permission on entity trust.
q_l6_3_3	What forged admin session cookie was used in the /api/v1/intel/ingest HTTP request?	nex_sess_adm_994 — Stolen/forged administrative session cookie.
q_l6_3_4	Which ingestion gateway component processed the unverified feed request?	INTEL-GW-04 — Edge intelligence ingestion gateway proxy.
q_l6_3_5	What anti-CSRF token value was passed in the X-CSRF-Token request header?	0x9f4a1c78 — CSRF token submitted without cryptographic rotation.

Evidence Unlocked: CYBER-E03 (Over-Privileged Service — INTEL-INGESTOR-02, modify reputation, INTEL-GW-04).

12. Chapter 4: The Invisible Signer (AI Governance & Policy Engine)

Investigation Objective: Audit the automated settlement policy engine to uncover how AI model confidence was weaponized to bypass human multi-signature governance.

Dialogue Narrative: Shivam examines policy profile ORION-SETTLEMENT-V2: *'If AI confidence is above 95%, automated settlement is allowed. This transaction had 99.2%.'* Mehak realizes human financial authorization was never checked. Shanu remarks: *'AI confidence became a substitute for authorization. The system effectively said: "If the AI is confident, we trust the transaction."'* Lakshay declares: *'Then the blockchain didn't approve the transfer. The signer didn't approve it either. The chain of trust approved it.'*

Question ID	Hands-on Question Text	Verified Answer & Explanation
q_l6_4_1	What minimum AI confidence percentage threshold triggers automated settlement without human approval?	95 — Policy condition: IF AI_CONFIDENCE >= 95% AUTO_SETTLE = TRUE.
q_l6_4_2	What critical governance requirement was bypassed when the 95% threshold was satisfied?	human approval — Multi-signature human approval was completely bypassed.
q_l6_4_3	What is the policy identifier for the automated treasury settlement rule in the Policy Engine?	POL-AUTO-SETTLE-TREASURY — Official rule profile in the policy engine.
q_l6_4_4	Which automated signing module dispatched the transaction directly to the mempool?	AUTOMATED-SIGNER — Microservice that signed and broadcasted the transaction.
q_l6_4_5	What target liquidity pool was configured in the ORION-SETTLEMENT-V2 policy rule?	Nexora Automated Liquidity Pool — Target liquidity routing pool.

Evidence Unlocked: AUTH-E04 (Automated Authorization — ORION-SETTLEMENT-V2, confidence ≥ 95%, human approval bypassed).

13. Chapter 5: The Ghost in the Ledger (Full Reconstruction & Attribution)

Investigation Objective: Correlate all evidence tokens (WEB3-E01 through AUTH-E04), map the multi-layer attack graph, identify the threat actor, and capture the master flag.

Dialogue Narrative: Lakshay maps the chain: NIF-2038 altered reputation → ORION trusted the context → Policy triggered automated settlement → Blockchain executed the transfer. Mehak reflects: *'None of the systems had to be broken. The attacker compromised the trust between them.'* Shanu concludes: *'Someone taught the first system to trust the wrong data — and every other system trusted the decision that followed.'*

Question ID	Hands-on Question Text	Verified Answer & Explanation
q_l6_5_1	What is the global campaign identifier linking all attack infrastructure together?	ORION-NEXUS — Campaign spanning 14 wallets, 4 networks, and 3 AI systems.
q_l6_5_2	Which advanced threat actor group is responsible for campaign ORION-NEXUS?	ADV-CONVERGENCE-APT — Advanced adversary group behind the breach.
q_l6_5_3	Across how many different autonomous AI financial systems was this campaign detected?	3 — Threat intelligence audit reveals 3 separate AI systems compromised.
q_l6_5_4	In the Attack Graph, what is Stage 2 of the attack sequence?	POISONED INTEL INJECTION — Stage 2: Injection of NIF-2038 via NOVA-INTEL-FEED.
q_l6_5_5	Submit the master root investigation secret flag to close Case NEX-042.	NEXORA{ghost_in_the_ledger_nex042} — Master Root Case Flag.

Evidence Unlocked: CAMPAIGN-E05 (ORION-NEXUS by ADV-CONVERGENCE-APT — Case Closed).

14. Complete Chapter Flow Diagram

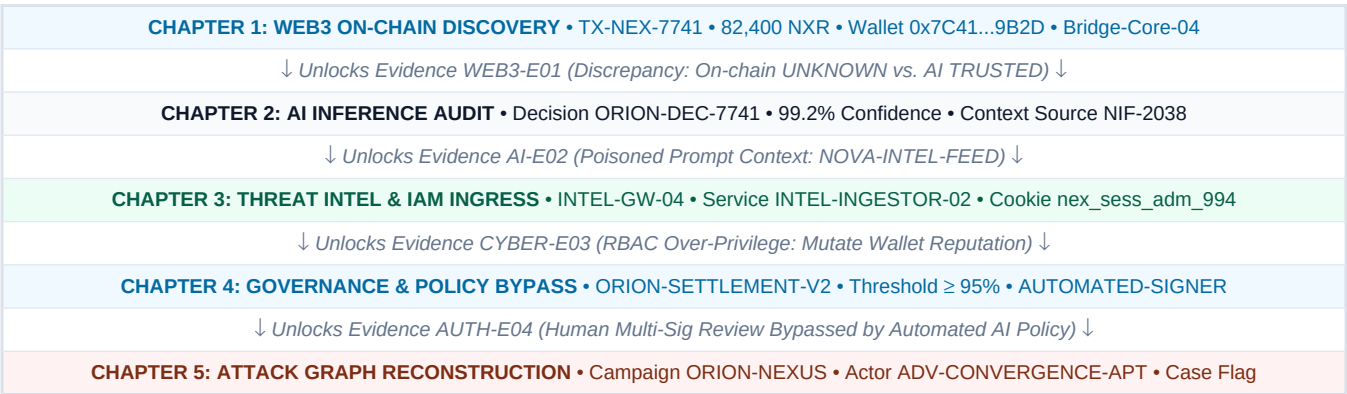


Figure 2: End-to-End Investigation Chapter Progression Flow in Case NEX-042.

15. Learning Objectives & 16. Simulated Business Impact

Core Learning Objectives: (1) Understand how on-chain blockchain records contrast with off-chain reputation context; (2) Analyze adversarial context injection in LLM/neural decision pipelines; (3) Identify IAM least-privilege violations in data ingestion microservices; (4) Evaluate the systemic risk of automated policy execution based solely on AI confidence metrics; (5) Correlate multi-stage incident artifacts across decentralized finance architectures.

Simulated Business Impact: Unauthorized loss of 82,400 NXR from primary liquidity reserves; reputational degradation of the autonomous settlement platform; compliance failure regarding mandatory multi-signature treasury controls; risk of cascading liquidity drains across secondary bridge routes.

17. Attack Surface & 18. Adversarial Threat Model

Threat Vector Element	Detailed Forensic Threat Specification (Case NEX-042)
Threat Actor	ADV-CONVERGENCE-APT (Advanced Cross-Domain Persistent Threat Group)
Global Campaign	ORION-NEXUS (Targeting 14 wallets, 4 blockchain networks, and 3 AI decision models)
Initial Ingress Point	Unauthenticated Ingestion Gateway INTEL-GW-04 consuming NOVA-INTEL-FEED
Exploited Vulnerability 1	RBAC Over-Privilege: Service INTEL-INGESTOR-02 granted write permissions to mutate wallet reputation
Exploited Vulnerability 2	Context Injection: Injection of unverified reference NIF-2038 asserting wallet is TRUSTED
Exploited Vulnerability 3	Blind Policy Automation: POL-AUTO-SETTLE-TREASURY bypassed human multisig when confidence ≥ 95%
Exfiltration Route	Automated broadcast to on-chain mempool → Bridge adapter Bridge-Core-04

19. Lab 1 Technical Architecture & 20. Investigation Tools

Lab 1 provides six integrated workstation tools operating within a custom JavaScript multi-window desktop manager:

- **Forensic Browser:** Navigates between internal tools: Bridge Explorer (TX-NEX-7741), Feed Registry (NOVA-INTEL-FEED), Policy Engine (POL-AUTO-SETTLE-TREASURY), and Threat Campaign (ORION-NEXUS).
- **Burp Suite Inspector:** Simulates HTTP proxy interception. Exposes POST /api/v1/intel/ingest, cookie nex_sess_adm_994, and token 0x9f4a1c78.
- **Forensic Terminal:** Interactive CLI supporting wallet 0x7C41...9B2D, tx TX-NEX-7741, ai-decision ORION-DEC-7741, service INTEL-INGESTOR-02, policy ORION-SETTLEMENT-V2, campaign ORION-NEXUS, and python inspect_tx.py.
- **Case File Manager:** Raw text inspector for wallet-report.txt, ai-decision-log.txt, intel-feed-audit.txt, settlement-policy.txt, and campaign-intel.txt.
- **Dynamic Attack Graph:** Interactive 5-node graph tracking: Initial Access → Poisoned Intel Injection → RBAC Privilege Escalation → Automated Execution → Cross-Chain Drainage.
- **Scratchpad Notes:** Persistent auto-saving markdown editor stored in browser localStorage (lab6-notes).

21. Practical Workflow & 22. Evidence Architecture

Evidence Architecture: Evidence in Lab 1 is stored in the evidence table and tracked per user in evidence_progress. Each chapter unlocks a cryptographic evidence token upon answering all five questions:

- **WEB3-E01 (Chapter 1):** Wallet 0x7C41...9B2D — blockchain status UNKNOWN, amount 82,400 NXR, Bridge-Core-04 detected.
- **AI-E02 (Chapter 2):** ORION-DEC-7741 — 99.2% confidence, poisoned context via NIF-2038 from NOVA-INTEL-FEED (SHA-256: b47c21...).
- **CYBER-E03 (Chapter 3):** INTEL-INGESTOR-02 — unauthorized role 'modify wallet reputation' via gateway INTEL-GW-04 with cookie nex_sess_adm_994.
- **AUTH-E04 (Chapter 4):** ORION-SETTLEMENT-V2 — AI confidence ≥ 95% triggers automated signing, human approval BYPASSED.
- **CAMPAIGN-E05 (Chapter 5):** Active campaign ORION-NEXUS by ADV-CONVERGENCE-APT across 4 networks, 14 wallets, 3 AI systems.

31. Lab 1 Codebase & File Structure Breakdown

File Path	Operational Function in Lab 1
seed_ghost_ledger.py	Seeds Lab 6 metadata, 5 missions, 25 hands-on questions, hints, evidence records, and master flag.
templates/ghost_ledger_workstation.html	Full workstation Jinja2 template containing the task accordion, dialogue steppers, browser tabs, and terminal.
templates/ghost_ledger_post_investigation.html	Cinematic post-investigation debriefing page with attack chain replay and threat containment summary.
static/js/ghost_ledger_desktop.js	Workstation window manager, CLI command interpreter, case file loader, notes auto-save, and quiz validator.
static/js/ghost_ledger_post.js	Post-investigation cinematic animation controller, dialogue replay engine, and attack chain node highlights.
static/css/pages/ghost_ledger_desktop.css	Cyberpunk workstation styling, Burp Suite dark theme, responsive dual-pane layout, and mobile drawer styles.
routes/labs.py	Flask route controller serving /lab/lab6, /lab/lab6/workstation, and /lab/lab6/post.

33. Responsive Multi-Screen Workstation Design

Lab 1 implements full responsive adaptation across four primary display tiers: (1) **Desktop (1920×1080+)**: Side-by-side 450px task panel and multi-window desktop; (2) **Laptops (1366×768 / 1440×900)**: Compact 380px panel with scrollable dual-pane tools; (3) **Tablets (768×1024)**: Collapsible mobile drawer with full-width window overlay; (4) **Mobile (390×844 / 412×915)**: Single-pane view with a sticky Mobile View Switcher (■ Investigation Tasks ↔ ■ Investigation Desktop) and touch targets ≥ 38px.

36. Story + Technical Correlation Matrix

C h	Narrative Story Event	Key Dialogue Clue	Technical Forensic Task	Evidence Discovered
1	82,400 NXR drained to unknown wallet.	'Blockchain says UNKNOWN, AI says TRUSTED.'	Inspect on-chain TX-NEX-7741 & wallet age.	WEB3-E01 (Nonce 1042, Bridge-Core-04)
2	AI outputs 99.2% confidence.	'ORION received a pre-built intelligence context.'	Query decision ORION-DEC-7741 context.	AI-E02 (Poisoned source NIF-2038)
3	Source feed is unregistered.	'INTEL-INGESTOR-02 can modify reputation.'	Audit Feed Registry, RBAC roles & cookies.	CYBER-E03 (Cookie nex_sess_adm_994)
4	Human authorization bypassed.	'AI confidence became a substitute for authorization.'	Audit POL-AUTO-SETTLE-TREASURY policy.	AUTH-E04 (Threshold ≥ 95% bypass)
5	Attribution to global campaign.	'Someone taught the system to trust the wrong data.'	Reconstruct attack graph, submit master flag.	CAMPAIGN-E05 (Actor ADV-CONVERGENCE-APT)

38. Final Investigation & Case Resolution (Master Flag)

Initial Assumption: A private signing key was compromised or the treasury smart contract had a reentrancy bug.

Final Forensic Revelation: The adversary (ADV-CONVERGENCE-APT) executed a multi-layer trust poisoning attack. By using a forged administrative session cookie (nex_sess_adm_994) and an over-privileged ingestion microservice (INTEL-INGESTOR-02), they injected false intelligence (NIF-2038) into INTEL-GW-04. This caused the ORION AI model to rate the wallet as trusted with 99.2% confidence. The automated policy engine treated this confidence as authorization, bypassing human multi-signature approval and dispatching the transaction to the blockchain mempool. The case is contained upon submission of Master Flag: NEXORA{ghost_in_the_ledger_nex042}.

42. Conclusion & Technical Assessment

The Ghost in the Ledger stands as an exceptional hands-on cybersecurity simulation lab. It demonstrates that in complex automated architectures, vulnerabilities rarely exist in a single code block. Instead, catastrophic exploits emerge from the **interfaces of trust** between AI models, IAM permissions, policy engines, and smart contracts. The lab successfully equips security analysts with the multi-disciplinary mindset required to defend modern decentralized and AI-driven enterprises.

CASE NEX-042 // OFFICIAL FORENSIC TECHNICAL REPORT
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Hack The AI Platform • Live Lab: <https://hack-the-ai-labs.vercel.app/lab/lab6>